

ORIGINAL RESEARCH

A deep neural network model with GCN and 3D convolutional network for short-term metro passenger flow forecasting

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Funding information

Guiding Project of Fujian Science and Technology
Plan, Grant/Award Number: 2021H0019; Natural
Science Foundation of Fujian Province, China,
Grant/Award Number: 2021J011202; National
Natural Science Foundation of China, Grant/Award
Numbers: 51305142, 51305143

Abstract

Rail transit has many advantages, such as large passenger capacity, convenience, safety, and environmental protection, making it the preferred travel mode for most passengers. Deep learning has become an effective method for short-term rail passenger flow prediction. A deep learning model which combines a graph convolutional network and a three-dimensional Convolutional Neural Network improved by a residual module and an attention mechanism (ARConv-CGN) is proposed. First, historical passenger inflow and outflow data are aggregated into three patterns: recent pattern, daily pattern, and weekly pattern separately. The GCN is applied to capture the spatiotemporal and topological information of passenger flows in each pattern. Second, a three-dimensional convolutional neural network is used to deeply integrate three patterns of passenger flow information. Additionally, a residual module is used to increase the number of neural network layers and prevent the gradient of the deep neural network from disappearing. Finally, an attention mechanism is also introduced to adjust the importance of passenger flow in different pattern, so that improving the performance of the model. Training with passenger flow data from automatic fare collection on weekdays in Beijing and Xiamen provides a good demonstration of the superiority of ARConv-CGN in metro passenger flow forecasting.

1 | INTRODUCTION

Nowadays, with the development of urban public transportation system, accurate traffic prediction is one of the most basic and critical tasks of urban traffic control and management. As the metro is more and more necessary in the urban public transportation system, the increasing passenger flow sets a higher request for the operation of the metro, and a little carelessness may affect the travel of passengers. In order to improve the service efficiency of the metro system, it is particularly important to grasp the dynamic trend of metro passenger flow in the short term. One of the basic problems is how to accurately predict the passenger flow of each station. The metro passenger volume prediction has attracted more and more attention from academia and related departments, and has become a research hotspot in the field of intelligent transportation systems [1, 2].

In earlier studies, the raw traffic state data for each time interval was usually converted into a vector or sequence, and then

a mathematical statistical model was used for prediction, such as autoregressive integrated moving average (ARIMA), hidden Markov model (HMM) and k-nearest neighbour (KNN) [3–5]. However, the aforementioned mathematical statistical model only focuses on a single feature in the historical data, and has high requirements for the stability of the time series. The prediction accuracy needs to be improved when facing nonlinear and complex traffic data.

With the development of machine learning, passenger flow prediction is no longer limited to the mining of a single feature. By extracting features that may affect the passenger flow situation, some machine learning-based models and hybrid forecasting models are introduced into passenger flow forecasting. Machine learning techniques are first used to extract features, and then perform prediction through the features [6]. Some studies combine ARIMA with wavelet decomposition, random forest and BP neural network for passenger flow prediction in urban rail transit [7–9]. In addition, the support vector machine [10, 11] model are also widely used in the research

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of passenger flow prediction. Although the above models can consider various spatiotemporal features, they generally rely on expert experience, and they are usually applicable to a single station rather than the entire urban rail transit network, which limits the accuracy improvement of passenger flow prediction.

Deep learning models have been shown to outperform traditional methods as a branch of machine learning in capturing complex correlations from big data and researchers have gradually attempted to apply models in these fields to passenger flow prediction as a branch of machine learning [12]. Given the good performance of time series forecasting, recurrent neural network (RNN) and their variants, such as long short-term memory (LSTM) [13–15] and gated recurrent unit (GRU), are widely used in mainstream research on traffic forecasting [16, 17]. However, RNN-based models only exploit the temporal features in travel behaviour, while ignoring the potential spatial correlations in the transportation network.

To take spatial correlations between stations into account even when they are far from each other, some researchers have modelled the passenger flow into grid points on two-dimensional (2D) images and used convolutional neural networks (CNNs) for forecasting [18, 19]. The residual network (ResNet) skip-connections between CNN layers to model the distant spatial correlations of stations by stacking adequate convolutional layers [20–22]. Despite the impressive results that have been achieved, these CNN-based models are designed for grid data and ignore the topology between metro stations.

In recent years, some researchers have modelled the data on the nodes of a graph and forecasted based on the graph, using graph neural networks (GNNs) for forecasting. These models can capture topological information between stations to deal with graph-structured data for traffic forecasting [23, 24]. DCRNN [25] regarded the traffic flow as a diffusion process on the graph and captures the spatial dependency with bidirectional random walks. STGCN [26] built a model with complete convolutional structures to capture the spatio-temporal features and it can train faster with fewer parameters. Some research combined GNNs with RNNs to capture spatial and temporal representations respectively [27–29]. Pan [30] used a dual-channel graph convolutional network to extract spatial features and used GRU to capture the temporal characteristics of the traffic series. He [31] used multiple graphs to encode the spatial-heterogeneous interstation correlations and combined an LSTM-encoder-decoder to complete the modelling of traffic flow data. Since attention mechanisms have been proposed [32], researchers have applied them to improve the performance of the model [33, 34]. Zhang [35] combined the residual network, graph convolutional network, and attention long short-term memory to forecast short-term passenger flow. Zhang [36] used a multi-scale attention network to capture multi-level temporal dynamics. Kong [37] proposed a graph talking-heads attention layer for capturing spatial dependencies and designed a gate temporal convolution layer for handling long temporal sequences.

However, there are three persistent problems that have been neglected.

1. Many methods model the spatial pattern and the temporal pattern separately, which can easily ignore their internal connection and does not consider the interaction between them, thus restricting the performance of the model.
2. Neural networks generally perform better with a stack of more layers, while GCN-based models generally only use one to four layers because GNNs benefit little from the depth and more layers may lead to inferior performance. This means that GCN-based models cannot go as deep as CNN-based models, making it difficult to obtain deeper and richer spatial features.
3. 2D CNNs cannot extract high-level features. Only a few research studies have focused on the application of three-dimensional (3D) CNNs that can utilize spatial-temporal features synchronously by 3D filters.

To overcome these shortcomings, in this paper, we propose a model called ARConv-GCN based on graph convolutional neural network (GCN) and 3D convolutional neural network combining with residual structure and attention mechanism (ARConv) for urban rail transit systematic short-term passenger flow forecasting. Finally, we evaluate and compare the performance of this model with several baseline models based on smart card data from Beijing metro and Xiamen metro. Experimental results show that our ARConv-GCN model outperforms those baseline models. The main work and innovations of this paper are as follows:

1. A deep learning architecture called ARConv-GCN is proposed, which combines GCN and 3DCNN for more accurate short-term passenger flow prediction in rail transit.
2. The residual structure is adopted in the model, which enables us to build a deep 3D network structure and use an attention mechanism to capture and global passenger flow trends, thereby improving the accuracy of short-term passenger flow prediction.
3. Experiments are conducted on two real datasets, and the results show that the proposed model outperformed other baseline passenger flow forecasting methods.

2 | METHODOLOGY

2.1 | Problem definition

Metro passenger forecasting is a basic spatiotemporal prediction problem. The purpose of this study is to use historical smart card data to predict the passenger flow of urban rail transit networks. Passenger data for each metro station is usually aggregated using a specific time interval to form a batch of time sequence. Define the urban rail transit network as a graph $G = (V, E, A)$ as shown in Figure 1, where V is the set of nodes and E denotes the set of edges. $|V| = n$ is the number of stations and $|E| = e$ is the number of connection

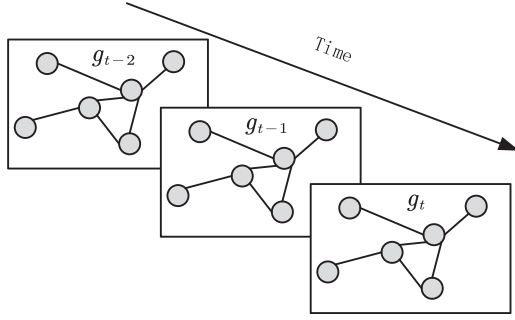


FIGURE 1 The time series traffic data

between stations. $A \in R^{n \times n}$ is an adjacency matrix which indicate a link between two stations exist or not. We treat the passenger flow volume as the attribute feature and the feature matrix $F = (X_{N,T}^r, X_{N,T}^d, X_{N,T}^w)$ is composed of recent passenger flow patterns, daily passenger flow patterns and weekly passenger flow patterns. The recent passenger flow patterns represent the passenger flow of the most recent previous periods of the forecast period, the daily passenger flow patterns represent the passenger flow of the same period of the previous day, and the weekly passenger flow patterns represent the passenger flow of the same period of the previous week. These three patterns are given by

$$X_{N,T}^p = \begin{pmatrix} X_{1,t}^p & X_{1,t-1}^p & X_{1,t-2}^p & \cdots & X_{1,t-m+1}^p \\ X_{2,t}^p & X_{2,t-1}^p & X_{2,t-2}^p & \cdots & X_{2,t-m+1}^p \\ \cdots & \cdots & \cdots & \ddots & \vdots \\ X_{n,t}^p & X_{n,t-1}^p & X_{n,t-2}^p & \cdots & X_{n,t-m+1}^p \end{pmatrix} \quad (1)$$

where p represents the overall of three passenger patterns, including recent, daily, and weekly patterns. m denotes the past several time steps used to predict. $X_{j,t}^p$ indicates the passenger flow vector of the j station in the k time interval in the p passenger flow patterns and each time interval will generate a feature matrix as input to our model. Therefore, the problem can be defined according to Equation (2), whereby f is the mapping function to be learnt using the proposed deep learning architecture.

$$X_{n,t+1} = f\left(A; X_{N,T}^r, X_{N,T}^d, X_{N,T}^w\right) \quad (2)$$

2.2 | Model architecture

The ARConv-GCN model architecture is shown in Figure 2. In this framework, the historical passenger inflow and outflow data are aggregated into three patterns: the recent pattern, the daily pattern, and the weekly pattern, separately. The GCN model is applied to capture spatiotemporal and topological information of passenger flows in each pattern. There are two graph

branches that dealt with the inflow and outflow data, and the outputs of them are merged and used as inputs to the attention residual 3D convolutional network (ARConv) to extract richer information. The ARConv layer is used to deeply fuse inflow and outflow information and synchronously extract high-level spatial and temporal features. Inside ARConv, residual structure can increase the number of neural network layers and prevent the gradient of the deep neural network from disappearing, thus extract deeper long-range dependencies that GCN cannot extract. The attention mechanism quantifies the importance of the extracted features and strengthens the extraction of global passenger flow features. After being flattened by the flattening layer, the output is sent to the fully connected layer which consists of the fully connected operation and the neuron inactivation operation for dimensionality reduction and capturing the nonlinear relationship between high-dimensional features and prediction results. Finally, the output of fully connected layer is reshaped into the shape of the target for which the model is to be trained.

2.3 | Graph convolution network

Some traditional traffic prediction models often regard the traffic network as a grid matrix, ignoring the influence of network topology on the prediction accuracy. In this section, we apply GCN to capture the topological relationships between stations in an urban rail transit network. GCN is a semi-supervised model that can deal with graph structures which is an advancement of CNN in the field of graph structures. In recent years, GCN have received extensive attention and GCN layers have also undergone significant improvements, such as spectral graph [38], first-order filters [39] and Chebyshev polynomial [40]. GCN has been widely utilized in various fields due to its powerful ability to extract spatial information. By determining the relationship between the nodes or links based on physical topologies, GCN can encode the topological structure of rail transit, thereby capturing the spatial dependence of passenger flow.

The GCN functions can then be defined by ref. [41] as shown in Equation (3):

$$X^{(l+1)} = \sigma\left(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} X^{(l)} W^{(l)} + b^{(l)}\right) \quad (3)$$

Where $A \in R^{n \times n}$ is the adjacency matrix based on a predefined physical topologies. $\tilde{A} = A + I$ and I is the identity matrix. $\tilde{D} \in R^{n \times n}$ is the normalized node degree matrix of $\tilde{A}$. $X^{(l)} \in R^{n \times m}$ is the feature matrix of the l layer and m represents the number of features of each of the node. $W^{(l)} \in R^{n \times m}$ is the weight matrix of the l layer. σ is the *ReLU* function and b is the bias vector.

Since the inflow and outflow are closely related to the passenger flow of recent, daily, and weekly periods, we have aggregated the historical passenger flows into three patterns to input into the GCN: the recent pattern, the daily pattern, and the weekly

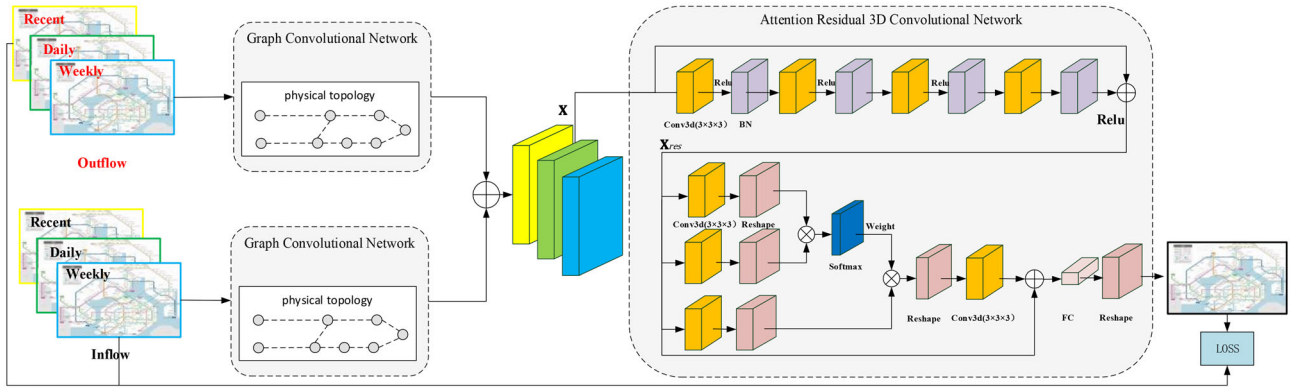


FIGURE 2 Model architecture

pattern with the same time steps. The passenger flow in adjacent periods will affect passenger flow in the next period. For example, when congestion occurs during peak hours on weekdays, passengers who cannot squeeze into the metro will have to wait for the next one. Since people's commute times are generally similar, it is necessary to predict passenger flows based on the previous day. The passenger flow will clearly reflect a trend on every weekday, such as the morning and evening peak. Due to a large number of commuters, passenger flows also present a certain regularity each week. For example, the passenger flow patterns of this Friday are similar to those of last Friday.

Their results are concatenated together before being input into attention residual 3D convolutional network (ARConv) to deeply integrate inflow and outflow information as shown in Figure 2 and the following Equation (4):

$$V = V_{\text{inflow}}^{l+1} \oplus V_{\text{outflow}}^{l+1} \quad (4)$$

where V is the input for ARConv, V_{inflow}^{l+1} and V_{outflow}^{l+1} are the output of the two GCN branches, $\oplus$ means stacking the two output tensors together according to the concatenation axis.

2.4 | 3D convolutional network

3D convolutions network have been widely used in computer vision, such as anomaly detection, video analysis, image classification, gesture recognition [42–44]. In rail transit data, the passenger flow of different stations in the same time interval is not independent, but correlated with each other. This pure spatial correlation can be captured by two-dimensional convolution. However, with the passage of time, the passenger flow conditions of different time periods will affect each other, and there is a correlation in the time dimension. Therefore, the model needs to learn the temporal and spatial characteristics of rail passenger flow at the same time. Related studies have demonstrated that 3D convolutions network have also been

shown to be effective in learning spatiotemporal features [48]. Therefore, we apply 3D convolutions network to model the spatiotemporal features, by aggregating the inflow and outflow information output by the GCN layer, in order to make full use of the potential spatiotemporal information in the passenger flow. The values in 3D CNN flow as shown in Equation (5).

$$V^{l+1} = \text{Conv3Do}(K \cdot V + \Delta) \quad (5)$$

where V is the input, K is the 3D filters, Δ is the bias, Conv3Do means the convolution operation including 3D convolution and 3D pooling. V^{l+1} is the high-level output feature. The values v at position (x, y, z) in the j th feature map of i layer can be obtained by following equation [42].

$$v_{ij}^{xyz} = \text{ReLU} \left(b_{ij} + \sum_m \sum_{p=0}^{P-1} \sum_{q=0}^{Q-1} \sum_{r=0}^{R-1} w_{ijm}^{pqr} v_{(i-1)m}^{(x+p)(y+q)(z+r)} \right) \quad (6)$$

where m is the feature map index in the $(i-1)$ th layer, w_{ijm}^{pqr} is the (p, q, r) th value of the 3D kernel connected to m th feature map in the $(i-1)$ th layer, (P, Q, R) is the dimension of 3D filters.

2.5 | Residual structure

Information loss seems inevitable due to convolution and pooling mechanisms in 3D convolutions network [45]. So we use residual module and attention mechanism to improve this phenomenon. Deeper neural networks are harder to train and they can extract more information, so this paper refers to the classical residual network [46] in order to improve the number of neural network layers and prevent vanishing gradient phenomenon. In order to solve the degradation problem caused by the increase of network depth, we use four normalization layers, four convolutional layers, and skip the connections of these layers through identity mapping. In this paper, the identity mapping is simply performed, adding neither extra parameters nor computational complexity. Their

outputs are added to the outputs of the stacked layers, making the network easier to optimize. The residual module takes the shallow features extracted by the 3D CNN module as input, and then Richer deep spatiotemporal feature information is extracted through convolutional layers and normalization layers.

2.6 | Attention mechanism

The attention mechanism in deep learning refers to the human's attentional thinking mode, which can use limited computing resources to extract higher-value information from data [47]. Since three patterns of historical passenger flow information have been inputted, they each have a different role in passenger flow prediction. By capturing the connection between the three patterns of passenger flow information and assigning their weights, the accuracy of the prediction can be further improved. We used an attention mechanism to calculate a context vector that can express the global trend of passenger flow status on the track in order to learn the importance of passenger flow information at each moment. From Figure 2, we can see that there is one addition operation and three convolution branches in the attention mechanism structure. There are two different paths in the convolution branch, we use two convolution branches constitute the traffic flow attention weight extraction part, we use another convolution branch for traffic flow feature extraction part. The 3D attention mechanism module can be abstracted as the Equations (7)–(10).

$$\delta' = \text{Softmax}((R \circ (\text{Conv3D} \circ v)) \otimes (R \circ (\text{Conv3D} \circ v))) \quad (7)$$

$$\text{ff}'' = R \circ (\text{Conv3D} \circ v) \quad (8)$$

$$\tilde{v} = \text{Conv3D} \circ (R \circ (v \delta' \otimes \delta'')) \quad (9)$$

$$y = \tilde{v} \oplus v \quad (10)$$

where $\text{Conv3D} \circ$ means the 3D convolution operation, $R \circ$ means the reshape operation, $\oplus$ means the matrix addition operation and $\otimes$ means the matrix multiplication operation.

2.7 | Loss function

The loss function is MSE, which is the average squared difference between the predicted data and the actual data and the calculation formula as shown in Equation (11).

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (11)$$

Where N is the number of passenger flow samples, y_i is the actual passenger flow value, $\hat{y}_i$ is the predicted passenger flow value.

3 | CASE ANALYSIS

3.1 | Research object and data description

Two datasets from Beijing Metro and Xiamen Metro are used in the experiments. The first dataset is MetroBJ [48], which is collected from February 29, 2016, to April 3, 2016 with 17 lines and 276 stations. The second dataset is MetroXM which is collected from December 1, 2019, to January 3, 2020 with 2 lines and 52 stations. These data is extracted from about 98 million pieces of AFC data in 25 working days for 5 consecutive weeks. We only use weekday data and the first four weeks were used to train the model and the last week were used to test the model. The AFC data mainly includes card number(TICKET_ID), travel date(TXN_DATA), card-swiping time(TXN_TIME), the id of stations(STATION_ID), the id of lines (LINE) and the type of card(KHTYPE) as shown in Table 1. These AFC records are aggregated at 10 min time intervals to form the passenger inflow data and outflow data separately.

3.2 | Evaluation method and criteria

We use root mean squared error (RMSE), mean average error (MAE) and weighted mean absolute percentage error(WMAPE) as the evaluation metrics as shown in Equations (12)–(14):

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (12)$$

$$\text{MSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (13)$$

$$\text{WMAPE} = \sum_{i=1}^N \left(\frac{y_i}{\sum_{j=1}^N y_j} \left| \frac{y_i - \hat{y}_i}{y_i} \right| \right) \quad (14)$$

Where N is the number of passenger flow samples, y_i is the actual passenger flow value, $\hat{y}_i$ is the predicted passenger flow value, $\sum_{j=1}^N y_j$ is the sum of the actual passenger flow values.

3.3 | Baseline

To illustrate the superiority of our model, we compared the performances with the baseline models as follows:

HA: Historical average model, which is used to predict the future passenger flow using the average value of historical passenger flow data [49].

ARIMA: Autoregressive integrated moving average model.

The lag order, the degree of difference, and the order of the moving average, are set as 2, 1, 0 after fine-tuning [50].

TABLE 1 The sample of AFC dataset

TICKET_ID	TXN_DATA	TXN_TIME	STATION_ID	LINE	KHTYPE
15**562	20191202	71390	11	1	E-card
46**234	20191211	33775	21	2	E-card
13**757	20191227	61059	26	2	E-card

LSTM: Long short term memory neural network. The LSTM model we use consists of one fully connected layer and two hidden layers [14]. The optimizer is Adam with a learning rate of 0.001. The batch size is 32. The inputs are inflow series in the last 20 time steps while outputs are the next time step time series.

CNN: We apply two CNN layers and a fully connected layers [17]. The two hidden layers have 32 and 64 filters with 3×3 size kernel. The optimizer is Adam with a learning rate of 0.001. Our inputs and outputs are the same as that of the LSTM model.

Conv-LSTM: It has one fully connected layer and two hidden layers with 32 and 64 filters [19] and the kernel size is 3×3 . The optimizer is Adam with a learning rate of 0.001.

3D CNN: We use the 3D CNN model and the filter number we set is 16, kernel size is 3 [48].

STGCN: A deep-learning architecture use graph convolution and 1D convolution to capture spatial dependencies and temporal correlations respectively, we directly use the original network structure [26].

DCRNN: A deep-learning architecture integrates graph convolution into a gated recurrent unit, we directly use the original network structure [25].

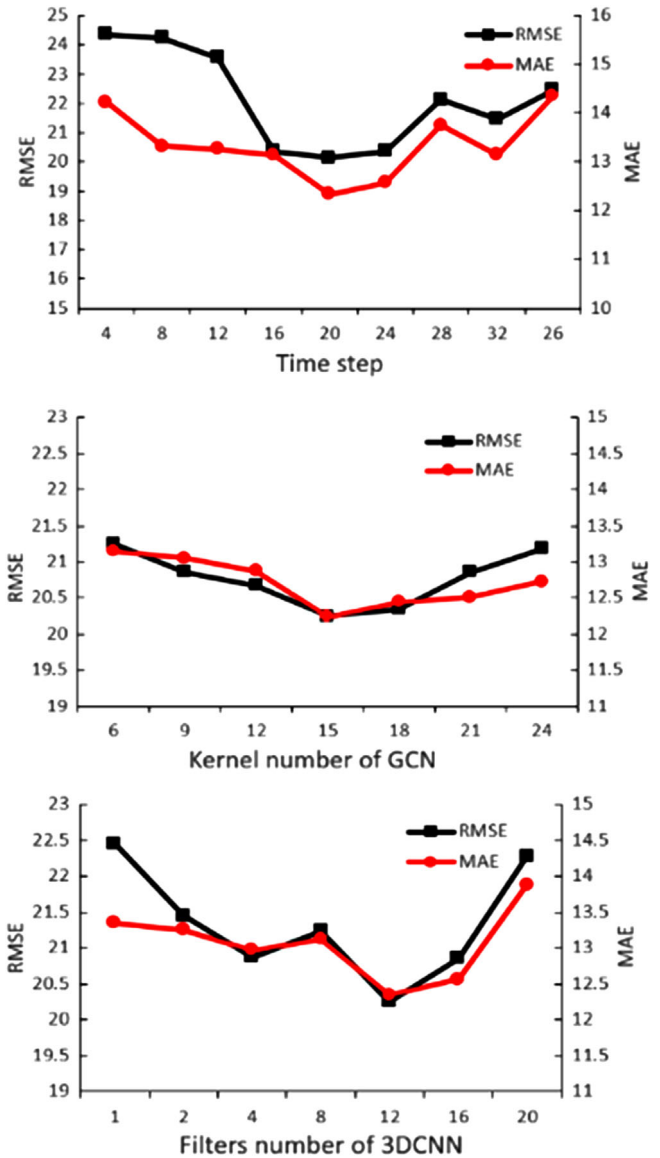
GCCN: A deep-learning architecture with multidimensional flows graph matrices, we directly use the original network structure [51].

Conv-GCN: A deep-learning architecture with GCN and 3D CNN for passenger flow forecasting, we directly use the original network structure [48].

ARConv-GCN-A: ARConv-GCN without attention mechanism.

ARConv-GCN-R: ARConv-GCN without residual module.

ARConv-GCN-R: ARConv-GCN without attention mechanism and residual module.

**FIGURE 3** Results of chosen hyperparameters

3.4 | Experimental settings

We used Keras and TensorFlow to implement our model. All models were implemented on the laptop with an Intel Core i7-10875H (16M Cache, up to 5.1 GHz), 16 GB of memory, and an NVIDIA GeForce GTX 2060. We trained our model using the Adam optimizer with a learning rate of 0.001. The training epoch was set to 200 and the batch size was set to 64 to train

the model. In total, there are three important hyperparameters in our model: time steps, the number of kernels in the GCN, and the number of filters in the 3DCNN. After many experimental tests and parameter adjustments, as shown in Figure 3, they were set to 20, 15, and 12, respectively.

TABLE 2 Comparison of different models in MetroBJ

MetroBJ	RMSE	MAE	WMAPE
HA	65.24	33.14	19.60%
ARIMA	50.54	27.42	15.53%
LSTM	38.19	21.99	12.71%
2D CNN	29.81	18.54	10.41%
Conv-LSTM	27.99	17.18	9.86%
3D CNN	27.17	16.92	9.73%
GCCN	26.98	16.52	9.68%
ST-GCN	26.53	16.30	9.62%
DCRNN	26.39	15.97	9.58%
Conv-GCN	25.69	15.61	9.53%
ARConv-GCN	23.03	14.45	9.12%
Improvement	10.35%	7.43%	4.30%

TABLE 3 Comparison of different models in MetroXM

MetroXM	RMSE	MAE	WMAPE
HA	58.26	30.59	16.5%
ARIMA	49.40	26.95	14.43
LSTM	35.98	17.76	10.88%
2D CNN	25.97	15.97	9.17%
Conv-LSTM	23.12	14.89	8.65%
3D CNN	22.89	14.71	8.55%
GCCN	22.48	13.57	8.35%
ST-GCN	21.83	13.26	8.03%
DCRNN	21.14	12.97	7.47%
Conv-GCN	20.83	12.88	7.54%
ARConv-GCN	17.97	11.85	7.16%
Improvement	12.85%	7.99%	5.03%

TABLE 4 Ablation study

Model	MetroBJ		MetroXM	
	RMSE	MAE	RMSE	MAE
ARConv-GCN-AR	26.02	15.91	21.25	13.19
ARConv-GCN-R	24.65	15.19	18.99	12.56
ARConv-GCN-A	24.18	14.88	18.24	12.10
ARConv-GCN	23.03	14.45	17.97	11.85

3.5 | Experimental results

Tables 2 and 3 show the performance of ARConv GCN compared with other baseline methods on the MetroBJ and MetroXM datasets. Table 4 is the ablation study of four variants of our model. The time interval of these experiments is 10 min,

and the evaluation criteria described in Section 3.2 are used for comparison. These experiment shows that ARConv-GCN has achieved the best results on both data sets.

3.6 | Analysis of experimental results

1. It can be seen that the deep learning model is far superior to the traditional model based on mathematical statistics. In the MetroBJ and MetroXM datasets, HA is the worst performing model, with RMSE of 65.24 and 58.26, and MSE of 33.14 and 30.59, respectively. ARIMA also performs poorly because it misses important spatial and topological information.
2. We further compare our model with the classic deep learning models LSTM and CNN. In deep learning models, LSTMs may not be able to model spatial correlations, and CNNs may not be able to capture temporal dynamics. In most cases, complex deep learning architectures such as ConvLSTM, STGCN, DCRNN, and GCNN have produced more favourable results than a single model. This is mainly due to the utilization of temporal and spatial features in these models.
3. In order to verify the influence of different components on ARConv-GCN, we conducted an ablation study as shown in Table 4, which proved that our improvement on 3DCNN is effective.
4. ARConv-GCN achieves the most accurate prediction in comparison to both traditional and deep learning models with the improvement RMSE of 10.35% and 12.85%, MAE of 7.43% and 7.99%, WMAPE of 4.30% and 5.03% for the two datasets, respectively.

AUTHOR CONTRIBUTIONS

Xuanrong Zhang: conceptualization, writing - original draft, data curation, formal analysis, investigation, methodology, software, validation, visualization, writing - review & editing; Cheng Wang: project administration, resources; Jianwei Chen: funding acquisition; Ding Chen: supervision.

ACKNOWLEDGMENTS

This work was financially supported by the Guiding Project of Fujian Science and Technology Plan (Grant No. 2021H0019). This work was financially supported Natural Science Foundation of Fujian Province, China (2021J011202). This work was financially supported by the National Natural Science Foundation of China (Grant Nos. 51305142, 51305143).

CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Research data are not shared.

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How to cite this article: Zhang, X., Wang, C., Chen, J., Chen, D.: A deep neural network model with GCN and 3D convolutional network for short-term metro passenger flow forecasting. *IET Intell. Transp. Syst.* 1–9 (2023). <https://doi.org/10.1049/itr2.12352>